01

MGU BCA - 3RD SEM

Feature Engineering

Module 1 - Introduction and Preprocessing

By Aben Sojan

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Complete question bank: every important, repeated and sure-mark question for this module, each with a full-mark model answer. Nothing important left out.

EXAM PATTERN (from real MGU question papers)

FE theory exam: 35 marks, 1 hour (MG3DSEBCA201 Honours model paper, mgu.ac.in Oct 2025).
Printed: 10 x Part A (2 marks) + 5 x Part B (5 marks) = 45 marks printed, so there IS choice.
Likely: Part A all compulsory (20) + answer any 3 of 5 in Part B (15) = 35. Exact choice rule UNVERIFIED - confirm from the paper.
Part A questions carry Bloom tags R/U/An and CO mappings in the paper.
Module 1 (data types, missing data, cleaning, scaling) gave 6 of the 10 Part A questions in the model paper - it is the Part A bank.

TOPIC 1

Data and Features - Introduction

1

Data is raw facts and figures collected from the real world, like student marks or house prices.

A feature is one measurable property of the data, like age, height or marks.

Each column in a dataset is usually one feature; each row is one record.

Good features make a machine learning model learn faster and predict better.

QUESTION BANK

1 What is a feature in machine learning? Give examples.

IMPORTANT

Definition: A feature is a single measurable property of the data that is given as input to a machine learning model.

In a dataset, each column is usually one feature and each row is one record or example.

The model learns patterns only from the features it is given, so the choice of features decides what the model can learn.

Examples: in a house-price dataset, area in square feet, number of bedrooms and location are features.

In a student dataset, age, attendance percentage and marks are features.

In a spam filter, features can be the number of links in the mail or the presence of words like lottery.

Features can be numerical (age, salary) or categorical (city, gender).

Raw columns are sometimes not enough, so new features are created from old ones; this is called feature engineering.

Example: from a date of birth column we can create an age feature that the model understands

better.

In short, features are the language through which data speaks to the model, and good features lead to good predictions.

2 What is data? How do features relate to a dataset?

IMPORTANT

Definition: Data is the raw collection of facts and figures gathered from the real world, like student marks or house prices.

A dataset organises this data as a table: each row is one record and each column is one feature.

So features are simply the columns of the dataset that describe each record.

Example: a student table has rows for each student and columns like name, age and marks.

Here age and marks are the features, because they are the measurable inputs the model will learn from.

Name is also a column, but it is usually not a useful feature since it carries no pattern.

The target column, like final result pass or fail, is what the model tries to predict and is not counted as a feature.

In short, data is the whole table, and features are the meaningful columns of that table that drive learning.

3 Differentiate between data and information.

IMPORTANT

Definition: Data is raw unprocessed facts and figures, while information is data that has been processed into something meaningful.

Data on its own answers nothing: a list of marks 45, 67, 89 tells you nothing until you know whose marks they are.

Information adds context and purpose: the class average is 67 and Aben scored 89, the highest in class.

Data is the input; information is the output of processing, cleaning and analysing that data.

Example: daily temperature readings for a year are data; the conclusion that July is the hottest month is information.

In machine learning, raw datasets are data, and the patterns the model learns from them become information.

One more difference: the same data can produce different information depending on the

question asked.

In short, data is raw material and information is the finished product made from it.

TOPIC 2

Importance of Features in Machine Learning

2

Models learn only from features, so bad features give bad predictions.

Good features improve accuracy without needing a bigger or costlier model.

Fewer useful features mean faster training and less overfitting.

In practice, feature engineering often matters more than the choice of algorithm.

QUESTION BANK

4 Why are features important in machine learning?

IMPORTANT

A machine learning model sees only the features, so the quality of features decides the quality of learning.

If the features are weak or irrelevant, even the best algorithm will give poor predictions.

Example: predicting house price without the location feature gives bad results, because location strongly affects price.

Good features improve accuracy without needing a bigger, slower or costlier model.

They also reduce training time, because the model finds patterns faster in clean, meaningful inputs.

Well-chosen features reduce overfitting, since the model does not get distracted by noise.

In real projects, improving features usually gives a bigger accuracy jump than tuning the algorithm.

Hence feature engineering is often called the most important step of the machine learning pipeline.

5 **'Feature engineering matters more than the algorithm.' Explain.**

IMPORTANT

This statement means that the information fed into a model matters more than the model itself.

A simple model with great features often beats a complex model with poor features.

Example: adding a days since last purchase feature improves a sales model more than switching algorithms.

The algorithm can only combine and weight what it is given; it cannot invent missing information.

If a key signal like customer location is absent, no algorithm can recover it.

On the other hand, with the right features, even basic models like linear regression perform very well.

That is why experienced data scientists spend most of their time on features, not on model tuning.

In short, features decide the ceiling of performance, while the algorithm only decides how close we get to it.

TOPIC 3

Data Types and Features

3

Numerical data is numbers. Discrete is countable (number of children); continuous is measurable (height).

Categorical data is labels. Nominal has no order (colour); ordinal is ordered (grades A, B, C).

Interval data has equal gaps but no true zero, like temperature in Celsius.

Ratio data has a true zero, like weight or height; you can say twice as heavy.

QUESTION BANK

6 Distinguish between discrete and continuous data with examples.

IMPORTANT

Definition: Discrete data takes separate, countable values, while continuous data can take any value in a range.

Discrete data is counted in whole numbers: number of students in a class, number of children in a family, number of cars sold.

It has clear gaps between values; you cannot have 2.5 children.

Continuous data is measured and can take decimal values: height 172.5 cm, weight 54.2 kg, time 9.58 seconds.

Between any two continuous values there is always another possible value.

Example: the number of books on a shelf is discrete, but the time taken to read one book is continuous.

This difference matters because some models and plots suit one type better than the other.

In short, countable values are discrete and measurable values are continuous.

7 Explain nominal and ordinal data with examples.

IMPORTANT

Both nominal and ordinal are types of categorical (label) data, but they differ in order.

Nominal data has categories with no natural order: gender, city, colour, blood group.

You can count how many fall in each category, but you cannot rank them.

Ordinal data has categories with a clear meaningful order: grades A, B, C, D or ranks 1st, 2nd, 3rd.

Example: shirt sizes S, M, L and XL are ordinal because M is clearly larger than S.

The order in ordinal data carries information, while in nominal data it does not.

This matters in encoding: ordinal data can use label encoding that respects order, nominal data usually needs one-hot encoding.

In short, nominal labels only name groups, while ordinal labels also rank them.

8 What is the difference between interval and ratio data?

IMPORTANT

Both interval and ratio are numerical data types, and the difference lies in the meaning of zero.

Interval data has equal gaps between values but no true zero: temperature in Celsius is the classic example.

0 degree Celsius does not mean no heat, it is just a point on the scale.

Ratio data has a true zero that means the complete absence of the quantity: weight, height, age, income.

Because of the true zero, ratios are meaningful for ratio data: 20 kg is truly twice as heavy as 10 kg.

But for interval data ratios mislead: 20 C is not twice as hot as 10 C.

Both types support addition and subtraction, but only ratio data supports meaningful multiplication and division.

In short, the true zero is what separates ratio data from interval data.

9 Why must categorical data be converted before using ML models?

IMPORTANT

Machine learning models do mathematics, and mathematics needs numbers, not words.

A model cannot multiply or compare the word Red, so every category must become a number first.

Encoding methods like one-hot encoding and label encoding perform this conversion.

Without conversion, most algorithms simply fail to run or silently misbehave on text columns.

Even models that accept categories internally still convert them to numbers behind the scenes.

The choice of encoding also matters: a wrong encoding can create false order between categories.

Example: encoding cities as 1, 2, 3 may trick a linear model into thinking city 3 is greater than city 1.

Hence converting categorical data correctly is a compulsory step before training.

10 What is the significance of ordinal data in decision making?

IMPORTANT

Ordinal data is categorical data with a meaningful order, like grades A, B, C or satisfaction levels.

Its significance is that the order itself carries information that pure labels do not.

Example: knowing a customer rated service as Poor, Average or Excellent lets a manager rank and prioritise complaints.

A decision maker can set thresholds on ordinal data: all feedback below Good needs attention.

Unlike nominal data, ordinal data supports comparisons like better than or worse than.

Unlike full numbers, it does not support exact maths: the gap between Good and Excellent is not measurable.

In practice it is often converted to ordered numbers (1 to 5) for analysis, as long as the order is respected.

In short, ordinal data matters because its built-in ranking directly guides prioritised decisions.

TOPIC 4

Handling Missing Data

4

Missing data means some values are absent, shown as NaN or blank cells.

Small gaps can be filled; columns that are mostly empty may be dropped.

Fill methods: mean, median, mode, or a fixed constant value.

Missing data must never be ignored silently; it can bias the model.

QUESTION BANK

11 What is missing data? How is it handled?

IMPORTANT

Definition: Missing data means some values are absent in the dataset, shown as blank cells or NaN.

It happens due to skipped survey answers, sensor failures or errors while merging files.

Missing data must never be ignored silently, because most models cannot train on empty cells and the gaps can bias results.

Handling method 1: fill small gaps using imputation, replacing blanks with the mean, median or mode of the column.

Handling method 2: fill with a sensible constant, like 0 for number of children or Unknown for a

city.

Handling method 3: drop the row or column, but only when very little data is affected or the column is mostly empty.

The right choice depends on how much is missing and why it is missing.

In short, detect the gaps first, then fill them sensibly or drop them carefully, and always document what was done.

12 Explain mean, median and mode imputation with examples.

IMPORTANT

Imputation means filling missing values with a sensible replacement instead of deleting the row.

Mean imputation replaces blanks with the average of the column; it suits normal numeric data without outliers.

Example: ages 20, 22, 24 with one missing become 20, 22, 24, 22, where 22 is the mean.

Median imputation replaces blanks with the middle value; it suits skewed data because outliers do not pull it.

Example: salaries 30000, 32000, 5000000 with one missing get the median 32000, not the distorted mean.

Mode imputation replaces blanks with the most frequent value; it is used for categorical columns.

Example: a city column missing in a few rows gets Mumbai if Mumbai appears most often.

In short: mean for normal numbers, median for skewed numbers, mode for categories.

13 When should a column with missing values be dropped instead of filled?

IMPORTANT

Filling is the default choice when only a small part of the column is missing, because the column still carries useful information.

A column should be dropped when most of its values are missing, commonly more than 50 to 60 percent.

Filling such a column would mean inventing more data than actually exists, which misleads the model.

Example: a middle name column missing in 80 percent of rows is better dropped than filled.

Similarly, drop a row only when it has too many missing values and very few rows are affected overall.

Dropping too many rows shrinks the dataset and can remove valuable patterns.

The decision should also consider importance: a highly predictive column deserves careful filling, not quick dropping.

In short, fill small gaps, drop columns that are mostly empty, and never drop blindly.

TOPIC 5

5

Data Cleaning

Data cleaning fixes errors so the model learns from correct data.

Common issues: duplicate rows, wrong formats, spelling mistakes, outliers.

Steps: remove duplicates, fix data types, correct invalid entries, handle outliers.

Cleaning is always the first step of any ML pipeline.

QUESTION BANK

14 **What is data cleaning? Explain its steps.**

IMPORTANT

Definition: Data cleaning is the process of fixing errors and inconsistencies in raw data so the model learns from correct data.

It is always the first step of any machine learning pipeline, because dirty data gives unreliable models.

Step 1: remove duplicate rows, which would otherwise give extra weight to repeated records.

Step 2: fix data types, converting values like text numbers into real numbers.

Step 3: correct invalid entries, such as negative ages or impossible dates.

Step 4: handle outliers, the extreme values that can distort the model.

Step 5: standardise text, making entries like Male, male and M into one form.

Clean data trains faster, gives higher accuracy and produces results that can be trusted.

15 List common data quality problems and how to fix them.

IMPORTANT

Data quality problems are the common defects found in raw real-world data, and each has a standard fix.

Duplicate rows: the same record appears many times; fix by dropping duplicates.

Wrong data types: age stored as text; fix by converting to the correct type.

Outliers: impossible or extreme values like age 250; fix by removing or clipping them.

Inconsistent text: Male, male and M meaning the same thing; fix by converting to one standard form.

Missing values: blank cells; fix by imputation or careful dropping.

Spelling mistakes and extra spaces: fix with stripping and standardisation functions.

Finding and fixing these issues is what data cleaning is all about.

TOPIC 6

Feature Scaling

6

Feature scaling brings all features into a similar range.

It is needed because salary in lakhs and age in tens confuse distance-based models.

Two main methods: standardization (z-score) and normalization (min-max).

KNN, SVM and neural networks need scaling; tree models do not.

QUESTION BANK

16 Why is feature scaling needed in machine learning?

IMPORTANT

Definition: Feature scaling brings all features into a similar numeric range before training.

It is needed because features often live on very different scales, like age 0 to 100 and salary 0 to 1000000.

Distance-based models measure gaps between points, so the large-range feature dominates the distance completely.

Example: in KNN, salary differences of lakhs would drown out age differences of tens, making age nearly useless.

Gradient-based models also train slowly and unstably when features have very different scales.

Scaling puts every feature on a similar scale, so each feature gets a fair weight in the model.

The two standard methods are normalization (min-max to 0-1) and standardization (z-score).

In short, scaling stops big numbers from bullying small numbers inside the model.

17 Which models need scaling and which do not? Why?

IMPORTANT

Models that need scaling are those based on distances or gradients: KNN, K-means, SVM, neural networks and PCA.

These models compare or combine feature values directly, so unequal ranges distort their maths.

Example: K-means clustering on unscaled age and income would cluster almost entirely on income.

Models that do not need scaling are tree-based: decision trees, random forests and gradient boosting.

Reason: trees split one feature at a time using thresholds, so the range of other features never matters.

A split like age greater than 30 works the same whether salary is in rupees or lakhs.

Still, scaling tree models does no harm; it is simply unnecessary.

In short, scale for distance and gradient models, skip it for trees.

TOPIC 7

Normalization and Transformation

7

Normalization (min-max scaling) squeezes values into the 0 to 1 range.

Standardization converts values to z-scores with mean 0 and standard deviation 1.

Transformation changes the shape of data, like log transform for skewed data.

Use normalization for a fixed range; standardization when data is roughly normal.

QUESTION BANK

18 Distinguish between normalization and standardization.

IMPORTANT

Both are feature scaling methods, but they rescale data in different ways.

Normalization, or min-max scaling, squeezes every value into a fixed range, usually 0 to 1.

It is done by subtracting the minimum and dividing by the range, so the smallest value becomes 0 and the largest becomes 1.

Standardization, or z-score scaling, centers values around mean 0 with standard deviation 1.

It is done by subtracting the mean and dividing by the standard deviation, so values show how many deviations they sit from the mean.

Normalization is bounded between 0 and 1, while standardization is unbounded and handles outliers more gracefully.

Use normalization when you need a fixed range, like for neural networks; use standardization when data is roughly normal or has outliers.

In short, normalization fixes the range, standardization fixes the distribution shape.

19 What are the formulas for min-max normalization and z-score standardization?

IMPORTANT

Min-max normalization rescales a value x into the 0 to 1 range using the column minimum and maximum.

Formula: x_new = (x - min) / (max - min).

Example: a mark of 80 with min 0 and max 100 becomes (80 - 0) / (100 - 0) = 0.8.

Every value in the column is transformed the same way, so order and relative gaps are preserved.

Z-score standardization converts x into standard deviations away from the mean.

Formula: z = (x - mean) / standard_deviation.

Example: a value 70 with mean 60 and standard deviation 10 becomes (70 - 60) / 10 = 1.0, meaning one deviation above average.

After standardization the column has mean 0 and standard deviation 1.

20 What is data transformation? Give an example.

IMPORTANT

Definition: Data transformation changes the shape of a feature distribution to make it more suitable for modeling.

It is used when data is skewed, meaning most values pile on one side with a long tail.

The log transform is the most common example: it compresses large values more than small ones.

Example: income data with a few very high earners becomes much more balanced after applying log.

Other transforms include square root for mild skew and square for left-skewed data.

Transformation helps models that assume roughly normal data, like linear regression.

It is usually applied after checking a histogram of the feature.

In short, transformation reshapes the data so the model sees a cleaner pattern.